# LOCAL EMBEDDABILITY VERSUS EMBEDDABILITY UNDER CHOICE, AND LOCAL ISOMORPHISM VERSUS ELEMENTARY EQUIVALENCE

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*Allowed standard prerequisites (editorial).*

- Konig's lemma for finitely branching infinite trees. Used inside the counted proof at lines 640-644. Produces an infinite branch in the tree of partial embeddings, whose union is the required embedding.
- First order sentences capturing the first n levels of an omega-tree up to isomorphism. Used inside the counted proof at lines 648-654. Converts elementary equivalence into local isomorphism by transporting the finite initial segments.
- The back-and-forth (Ehrenfeucht-Fraisse) argument for elementary equivalence. Used inside the counted proof at lines 656-659. Proves the converse direction, with the player's strategy built from the given local isomorphisms.

**Lemma 0.1.** *(i) Assuming the axiom of choice, for any two $\omega$-trees with finite levels, $T_1$ is locally embeddable in $T_2$ if and only if it is embeddable in $T_2$.*

*(ii) Assuming the axiom of choice, $T_1$ and $T_2$ are locally isomorphic if and only if they are isomorphic.*

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*(iii) $T_1$ and $T_2$ are locally isomorphic if and only if they are elementarily equivalent.*

*Proof.* (i) Let $T_1$ be locally embeddable in $T_2$. Let $P_n$ be the family of all embeddings of the union of the first $n$ levels of $T_1$ into the union of the first $n$ levels of $T_2$. Let $P = \bigcup_{n\in\omega} P_n$ be partially ordered by extension. Then $P$ is a tree with $n$th level $P_n$, each of which is finite, and by assumption non-empty. By König's Lemma, $P$ has an infinite branch, and the union of this branch provides an embedding of $T_1$ into $T_2$.

(ii) This follows by the same proof.

(iii) Assume first that $T_1$ and $T_2$ are elementarily equivalent. For any $n$ there is a first order sentence capturing the first $n$ levels of $T_1$ up to isomorphism. More precisely, if $N$ is the number of nodes in these levels (clearly a finite number), the sentence says that there exist $N$ points, which are ordered in the correct way, and any other $n$ points which are linearly ordered and form an initial segment are contained in the $N$ first mentioned. Since $T_1$ and $T_2$ are elementarily equivalent, there must be corresponding $N$ points of $T_2$, which are ordered in the same way as those in $T_1$, and which form the union of the first $n$ levels. Hence the unions of the first $n$ levels of $T_1$ and $T_2$ are isomorphic.

Conversely, if $T_1$ and $T_2$ are locally isomorphic, then we can use back-and-forth (an Ehrenfeucht-Fraïssé game) to see that they are elementarily equivalent. Suppose that this has $n$ moves. Then player II has a winning strategy, in which he uses given isomorphisms between the unions of finitely many levels to decide what to do next. □